Natural deduction inference rules

Axiom:

$$Ax \quad \frac{}{\Gamma, p \vdash p}$$

Introduction of *negation*:

$$\neg I \quad \frac{\Gamma, p \vdash \bot}{\Gamma \vdash \neg p}$$

Elimination of *negation*:

$$\neg E \quad \frac{\Gamma \vdash p \qquad \Gamma \vdash \neg p}{\Gamma \vdash \bot}$$

Ab absurdo :

$$Abs \quad \frac{\Gamma, \neg p \vdash \bot}{\Gamma \vdash p}$$

Elimination of the false (*ex falso*):

$$\bot \; E \quad \frac{\Gamma \vdash \bot}{\Gamma \vdash p}$$

Introduction of *implication*:

$$\rightarrow I \quad \frac{\Gamma, p \vdash q}{\Gamma \vdash p \rightarrow q}$$

Elimination of *implication* (*modus ponens*):

$$\rightarrow E \quad \frac{\Gamma \vdash p \rightarrow q \qquad \Gamma \vdash p}{\Gamma \vdash q}$$

Introduction of *and*:

$$\wedge I \quad \frac{\Gamma \vdash p \qquad \Gamma \vdash q}{\Gamma \vdash p \wedge q}$$

Elimination of *and* on the left (note: we keep the left side):

$$\wedge E_g \quad \frac{\Gamma \vdash p \wedge q}{\Gamma \vdash p}$$

Elimination of *and* on the right (note: we keep the right side):

$$\wedge E_d \quad \frac{\Gamma \vdash p \wedge q}{\Gamma \vdash q}$$

Introduction of *or* on the left (note: we introduce it on the right):

$$\vee I_g \quad \frac{\Gamma \vdash p}{\Gamma \vdash p \vee q}$$

Introduction of *or* on the right (note: we introduce on the left):

$$\vee I_d \quad \frac{\Gamma \vdash q}{\Gamma \vdash p \vee q}$$

Elimination of *or*:

$$\vee E \quad \frac{\Gamma \vdash p \vee q \qquad \Gamma, p \vdash r \qquad \Gamma, q \vdash r}{\Gamma \vdash r}$$

Introduction of the *universal quantifier*:

$$\forall I \quad \frac{\Gamma \vdash F}{\Gamma \vdash \forall x \; F} \quad \text{if x is not free in } \Gamma$$

Elimination of the *universal quantifier*:

$$\forall E \quad \frac{\Gamma \vdash \forall x \; F(x)}{\Gamma \vdash F[t/x]}$$

Introduction of the *existential quantifier*:

$$\exists I \quad \frac{\Gamma \vdash F(t)}{\Gamma \vdash \exists x \; F(x)}$$

Elimination of the existential quantifier:

$$\exists E \quad \frac{\Gamma \vdash \exists x \; F \qquad \Gamma \vdash F \rightarrow G}{\Gamma \vdash G} \quad \text{if x is not free in } \Gamma \text{ and } G$$